Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

III SEMESTER B. Tech. (DSE)
END SEMESTER EXAMINATIONS
NOVEMBER 2023

SUBJECT: DATA STRUCTURES**(DSE-2122)**

(05 /12/2023)

Time: MAX. MARKS: 50

Instructions to Candidates:

- Answer **ALL** questions.
- Missing data may be suitably assumed.

1	Imagine a scenario where you are an urban planner tasked with optimizing the construction of roads to connect several neighbourhoods in a growing city. Each neighbourhood represents a vertex in a graph, and the roads between them have different associated costs. Let's consider a city with 7 neighborhoods (A, B, C, D, E, F, G) and their associated road construction costs: Table :-1 Source | Destination | Costs($) A | B | 4 A | C | 7 A | E | 3 A | F | 6 B | C | 8 B | D | 9 C | D | 10 C | E | 5 D | E | 6 D | F | 11 E | F | 12 E | G | 5 F | G | 10 Your goal is to minimize the overall cost of constructing roads while ensuring that all neighborhoods are interconnected. i) Describe how you would apply the principles of a minimum spanning tree using Kruskal's algorithm to efficiently plan the road network for the city. ii) Find the cost of construction.	5	CO4	BL4	PO1,2,3

<table>
<tr><td>2</td><td>i) Construct the binary search tree and threaded tree for the given data: BRAINCOMPUTERINTERFACE
ii) How many null pointers are present in each representation?</td><td>3</td><td>CO3</td><td>BL3</td><td>PO1,2,3</td></tr>
<tr><td>3</td><td>Consider the scenario: Joe is having three descendants Alo, Rio, Sang. Rio have two descendants Mark and Mary. Represent this data using tree data structure. Convert this tree into binary tree. Show each step with neat diagram.</td><td>2</td><td>CO3</td><td>BL3</td><td>PO1,2,3</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>4</td><td>Consider the scenario wherein there is a need to process a large dataset of student records for a university admission system. The dataset consists of various student attributes such as Student ID and test scores. Sort this dataset given below in Table 1 in descending order based on a specific attribute, say the students' test scores, to streamline the admission process.
Table :-1
<table>
<tr><td>Student ID</td><td>101</td><td>102</td><td>103</td><td>104</td><td>105</td><td>106</td><td>107</td><td>108</td><td>109</td><td>110</td></tr>
<tr><td>Test score</td><td>85</td><td>72</td><td>91</td><td>68</td><td>77</td><td>94</td><td>80</td><td>63</td><td>88</td><td>75</td></tr>
</table>
i) Explain how you would use the heap sort algorithm to efficiently sort the student records based on their test scores.
ii) Discuss the step-by-step process of heap sort and how it could be applied to sort the student records. Write all intermediate steps in detail with formula along with the intermediate tree and array representation.</td><td>5</td><td>CO4</td><td>BL4</td><td>PO1,2,3</td></tr>
<tr><td>5</td><td>Construct a binary tree from the given preorder traversal as CABDIHKMEFGJLNO and inorder traversal as AIDBKHMCFEJNLOG.
i) Draw the tree with all intermediate steps.
ii) Write its postorder traversal.</td><td>3</td><td>CO2</td><td>BL3</td><td>PO1,2,3</td></tr>
<tr><td>6</td><td>If the values of A, B, C, and D are 2, 3, 4, and 5 respectively, calculate the value of the following prefix and postfix expressions:
i) + – * A B C D
ii) A B * C – D +</td><td>2</td><td>CO2</td><td>BL3</td><td>PO1,2,3</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>7</td><td>Construct a B tree (right biased) of order 4 by inserting the following keys :- OBJECTRINDPGAMQ
i) From the constructed tree, delete the following keys: B, D, A, R, J one by one. Explain which type of target node is deleted and mention which method is applied to solve the underflow of b-tree. Write all the intermediate steps and tree representations for insertion and deletion operations clearly and neatly.
ii) What is meant by reflow state in b-tree?</td><td>5</td><td>CO4</td><td>BL4</td><td>PO1,2,3</td></tr>
<tr><td>8</td><td>Write the pseudocode for Merge sort and trace the algorithm for the input: 100,90,80,70,60,50.</td><td>3</td><td>CO1</td><td>BL3</td><td>PO1,2,3</td></tr>
<tr><td>9</td><td>Write a recursive user defined function to find Fibonacci series of the number n.</td><td>2</td><td>CO1</td><td>BL2</td><td>PO1,2,3</td></tr>
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10	Draw the Recursive Tree for the Towers of Hanoi problem where the number of the disks is four. The function call is towers (4, N, M, L,) where 4 = no.of disks, N=Source, M=destination, L=Auxiliary at the initial step. Mention all the moves from the constructed tree and find total no.of moves.	5	CO2	BL3	PO1,2,3
11	i) Construct an expression tree for the given expression: A+(B-C*(D/E)+F+G)*H^K-L. ii) Write the preorder and postorder traversal of the constructed tree.	3	CO2	BL3	PO1,2,3
12	What are the two ways of deleting a node from a binary search tree which contains left and right subtree?	2	CO1	BL2	PO1,2,3
13	i) Write the standard algorithm steps for converting infix to Polish expression. ii) Convert the given expression into Polish notation using standard algorithm. A+B-(L^M)*(P+O)-Q/S^R*T+Z Write all the intermediate steps, show the stack content and resultant expression at each step .	5	CO3	BL3	PO1,2,3
14	Write the algorithm for performing enqueue and dequeue operation on circular queue by considering all necessary conditions.	3	CO2	BL2	PO1,2,3
15	Compare and contrast singly and doubly linked list.	2	CO2	BL1	PO1,2,3